

Sample test 3

1. **Questions 1 and 2 refer to the following setup.** We are interested in p , the proportion of all people who are currently happy with their cell phone plans. In a small study done in 2012, it was found that in of a sample 150 people, there were 90 who were happy with their cell phone plans. Construct a 95% confidence interval for p .
2. How large should our sample be to estimate p within a 4% margin of error at a 95% confidence level?
3. Recent incidents of food contamination have caused great concern among consumers. A report showed that 35 of 80 randomly selected Perdue brand broilers tested positively for either campylobacter or salmonella, the leading bacterial causes of food-borne disease, whereas 66 of 160 Tyson brand broilers tested positive. Does it appear that the true proportion of non-contaminated Perdue broilers differs from that for the Tyson rand? Find the p-value.
4. 10 people were given one day crash course in reading. Each of 10 adults was given a reading test before the course and another test after the course. The results are shown below.

Adult	1	2	3	4	5	6	7	8	9	10
Score after course	48	42	43	34	50	30	43	38	41	38
Score before course	31	34	18	30	44	28	34	33	27	32

For this matched pairs experiment, what is the observed value of the test statistic in order to test the hypothesis that taking the course improves test scores?

5. Summary statistics computed for two independent samples are: $n_1 = 25$, $\bar{x}_1 = 7.30$, $s_1 = 1.05$, $n_2 = 25$, $\bar{x}_2 = 6.80$, and $s_2 = 1.20$.
The estimated standard error (ese) of the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is equal to (assume normality).

Questions 5 and 6 refer to the following setup. A manager wishes to estimate the relationship between the value of inventory items and their annual turnover rate in order to predict turnover rate. The following information was obtained from 6 items.

Value of item, x	1	1	2	2	3	4
Turnover rate, y	2	2	3	5	6	7

It is calculated that $\sum x = 13$, $\sum y = 25$, $\sum x^2 = 35$, and $\sum y^2 = 127$.

6. Find the slope of the regression line.
7. Find the predicted number of turnovers per year for an item valued at \$1.5.

Questions 8 and 9 refer to the following setup. A restaurant owner wants to determine whether customers who sit by the window and customers who sit in a central location of the restaurant differ in the average length of time they stay at their table. A random sample was taken of 50 customers who had a table by the window, and the mean stay time and standard deviation were calculated to be 65 minutes and 15 minutes, respectively. A random sample of 45 customers who sat in a central location had a mean and standard deviation of 55 minutes and 10 minutes, respectively. Conduct a hypothesis test at significance level $\alpha = 0.05$ to answer the restaurant owner's question.

8. Find the p-value for this hypothesis test.
9. Find a 95% confidence interval for the difference in stay time between customers who sit by the window and customers who sit in a central location of the restaurant.
10. Short Answer: Two random samples of the annual salaries of females and males were obtained. The samples of females and male yield $n_1 = 5$, $\bar{x}_1 = 43.9$, $s_1 = 5.4$ and $n_2 = 7$, $\bar{x}_2 = 42.8$, $s_2 = 2.7$, respectively. Let μ_1 and μ_2 denote the true mean salaries of the two populations (female and male salaries). The populations are normally distributed.
 - (a) Find the estimated value of $\mu_1 - \mu_2$ and the estimated standard error (ese).
 - (b) Construct a 98% confidence interval for $\mu_1 - \mu_2$.
 - (c) Is there evidence that salaries for females are on average higher than salaries for males? State the null and alternative hypotheses that you would use.
 - (d) Find the observed result of the test statistics.
 - (e) Find the p-value.
 - (f) Draw the conclusion.

Answers:

1. (0.5216, 06784)
2. 577
3. 0.7114
4. $t_{obs} = 4.29$, $\nu = 9$
5. .32
6. 1.73
7. 3.01
8. $Z_{obs} = 3.86$ and p-value = 0
9. (4.918, 15.082)
10. (a) $\bar{x}_1 - \bar{x}_2 = 1.1$, $ese = 2.62$ (b) $(-7.72, 9.92)$
 (c) $H_0: \mu_1 - \mu_2 = 0$, $H_1: \mu_1 - \mu_2 > 0$
 (d) $t_{obs} = 0.4196$ (e) $p - value > 0.1$
 (f) Since $p - value > 0.1 > \alpha = 0.05$, we do not reject H_0